

ABHINAV REDDY TERUPALLY

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SUMMARY

Applied AI / Data Science graduate student with hands-on experience building Python-based ML, deep learning, and LLM systems across NLP, RAG, forecasting, classification, and analytics. Strong in production-minded pipeline development, model evaluation, API integration, JSON-first workflows, and reliable AI systems using LangChain, FastAPI, scikit-learn, PyTorch, TensorFlow, SQL, and cloud/backend tools.

TECHNICAL SKILLS

Programming & Data: Python, SQL, R, C, JSON, pandas, NumPy

Machine Learning / Deep Learning: scikit-learn, XGBoost, PyTorch, TensorFlow, Hugging Face, Transformers, Regression, Classification, Clustering, Time-Series Forecasting, Anomaly Detection, Model Evaluation

GenAI / LLM Systems: LangChain, LangGraph, OpenAI API, Claude, Amazon Bedrock, ChromaDB, FAISS, Ollama, RAG, Prompt Engineering, Structured Outputs, Guardrails

APIs / Backend / MLOps: FastAPI, REST APIs, Pydantic, Streamlit, Docker, Airflow, MLflow

Cloud / Databases: AWS, Azure, GCP, PostgreSQL, SQLite, Oracle SQL, Spark

Developer Tools / Analytics: Git, GitHub, VS Code, Cursor, Power BI, Excel

EDUCATION

University of Delaware - Master of Science in Data Science

Aug 2024 - May 2026

Relevant Coursework: Natural Language Processing, Machine Learning, Analysis of Algorithms, Database Systems, Big Data Technologies, Mathematical Techniques in Data Science

Sastra University, India - Bachelor of Technology, Computer Science and Engineering

Jun 2019 - Jun 2023

Relevant Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Software Engineering, Machine Learning

EXPERIENCE

Research Assistant - University of Delaware

Sep 2025 - Present

- Build an end-to-end NLP/LLM pipeline in Python and LangChain to convert Stack Overflow posts into auditable sentence-level Entity-Aspect-Sentiment triples for downstream analysis.
- Design JSON-first workflows for entity extraction, mention tracking, coreference resolution, aspect classification, and triple assembly with strict parsing, schema validation, and traceable outputs.
- Improve pipeline robustness through guardrails, retries/backoff, deduplication, error analysis, and OpenAI model benchmarking to support reliable AI system evaluation.
- Document methodology, edge cases, and annotation comparisons clearly for research discussions and cross-functional collaboration.

Data Science Intern - Edureka, India

Aug 2023 - Aug 2024

Climate Change & Economy | Python, Random Forest, LSTM, Time-Series Forecasting

- Built a global country-year ML prototype for climate-policy analysis, integrating economic, emissions, agriculture, and carbon pricing datasets for regression, classification, clustering, and forecasting.
- Trained an LSTM forecasting model that reduced test error by ~40% for next-year carbon pricing revenue prediction.
- Applied Random Forest anomaly detection (AUC $\approx$ 0.80), crop production modeling ($R^2 = 0.93$), and K-means clustering to surface interpretable policy and economic patterns.

PROJECTS

Sanjaya AI - AI Advising Platform | Python, FastAPI, Next.js, LangGraph, ChromaDB, RAG

Feb 2025 - Present

- Built a full-stack AI advising platform that transforms student profiles into explainable, prerequisite-safe semester roadmaps using LangGraph and hybrid RAG.
- Implemented deterministic validation for prerequisite closure, term-aware scheduling, credit limits, and Pydantic schema checks to ensure feasible recommendations outside LLM generation.
- Develop citation-backed retrieval, conversational intake, and advisor-style Q&A to deliver clear, stakeholder-friendly recommendations.

E-Commerce Chatbot with Multi-Intent Classification | LangChain, Semantic Router, SQLite, OpenAI, Streamlit

Aug 2025 - Sep 2025

- Designed an AI chatbot with dual intent routing (FAQ vs SQL) using semantic-router and all-MiniLM-L6-v2, achieving 82.5% routing accuracy and 0.80 Macro-F1.
- Built OpenAI-powered NL-to-SQL search over a SQLite product catalog, enabling plain-English queries with clickable result links.
- Deployed a Streamlit app with Selenium scraping, CSV-to-SQL ingestion, and fallback handling for unmatched intents and context-length failures; reached 90% FAQ accuracy on paraphrase tests.

News Summarization Model Comparison | Python, Fine-tuning, T5, GPT-2

Mar 2024 - May 2025

- Prepared an abstractive summarization dataset pipeline for 55K Inshorts article-summary pairs (cleaning, tokenization, input formatting).
- Fine-tuned T5-small and GPT-2 on GPU, including a masked-label setup for decoder-only training.
- Compared models using ROUGE-1/2/L and BERTScore: T5 R-1 0.5365 / BERT 0.9061, GPT-2 R-1 0.2217 / BERT 0.8773.

Image Classification Pipeline with EfficientNet | Python, PyTorch, Transfer Learning

Nov 2024 - Dec 2024

- Trained an anime face recognition system using EfficientNet transfer learning for multi-class classification.
- Strengthened generalization with augmentation (random flips/rotations/crops) and regularization (dropout, early stopping).
- Achieved 92.4% validation accuracy, outperforming a baseline CNN by 6.1%.

Credit Card Launch Impact Analysis | Python, SQL, A/B Testing, Z-test

Oct 2024

- Analyzed 500K+ transaction records and 1,000 customer profiles to segment users for a new credit card launch.
- Designed a 100-customer A/B pilot focused on the 18–25 segment to evaluate targeting and offer strategy.
- Validated impact via Z-test hypothesis testing, showing a statistically significant ~40% lift in adoption and transaction value.